

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	1 October 2026
Team ID	SWTID-2026-4204
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Problem Statement:

Manually classifying support tickets into the correct category, priority, assignment group, and workflow can be time-consuming and inconsistent. The goal is to automate ticket classification using Flow Designer so that incoming tickets are identified and routed to the appropriate team with less manual effort.

Target Outcome:

Automatically classify incoming tickets based on their subject, description, and relevant ticket information, then trigger the appropriate workflow and assignment.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Brainstormed Ideas:

1. Classify tickets by category such as Incident, Request, Access, Billing, or Technical Issue.
2. Automatically assign priority based on keywords, impact, and urgency.

Idea Prioritization:

1. Automatic ticket classification and category assignment — High Impact / High Feasibility
2. Automatic routing to the correct assignment group — High Impact / High Feasibility
3. Priority assignment using urgency and impact — High Impact / Medium Feasibility
4. Automated notifications — Medium Impact / High Feasibility
5. Fallback/manual review for uncertain tickets — Medium Impact / High Feasibility
6. Predefined rules for common ticket types — Medium Impact / High Feasibility
7. Classification monitoring and reporting — Medium Impact / Medium Feasibility

Selected Idea:

Build a Flow Designer-based automation that classifies incoming tickets, assigns the appropriate category and priority, routes the ticket to the correct support group, and sends notifications. Unclear tickets should be sent through a fallback/manual review path.

3. Route classified tickets to the correct assignment group or support team.
4. Use Flow Designer conditions and actions to trigger classification and routing.
5. Send automatic notifications to the assigned team and ticket requester.
6. Apply predefined rules for frequently occurring ticket types.
7. Create a fallback path for tickets that cannot be classified confidently.
8. Track classification results and identify common ticket categories for process improvement.

Grouped Ideas:

- Classification: category detection, ticket type identification, predefined rules.
- Prioritization: urgency and impact based priority assignment.
- Automation & Routing: Flow Designer actions, assignment groups, workflow triggers.
- Communication: requester updates and team notifications.
- Exception Handling: manual review/fallback for uncertain classifications.
- Monitoring: classification tracking and continuous process improvement.



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

 10 minutes to prepare
 1 hour to collaborate
 2-8 people recommended



Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

 10 minutes

A

Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B

Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

C

Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) →

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

 5 minutes

PROBLEM

How might we [your problem statement]?



Key rules of brainstorming

To run a smooth and productive session

 Stay in topic.

 Encourage wild ideas.

 Defer judgment.

 Listen to others.

 Go for volume.

 If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP

You can select a sticky note and hit the pencil (switch to sketch) icon to start drawing!

Amar



Yuktesh



Person 3



Person 4



Person 5



Person 6



Person 7



Person 8



3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

Person 4

TIP

Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mind.

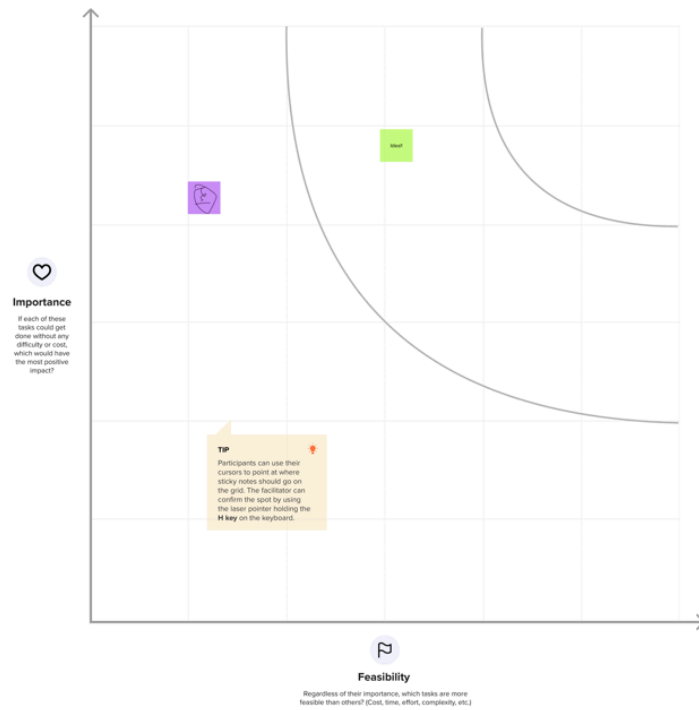
Step-3: Idea Prioritization

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes



Ideation Phase

Define the Problem Statements

Date	1 October 2026
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Maximum Marks	2 Marks

Customer Problem Statement Template:

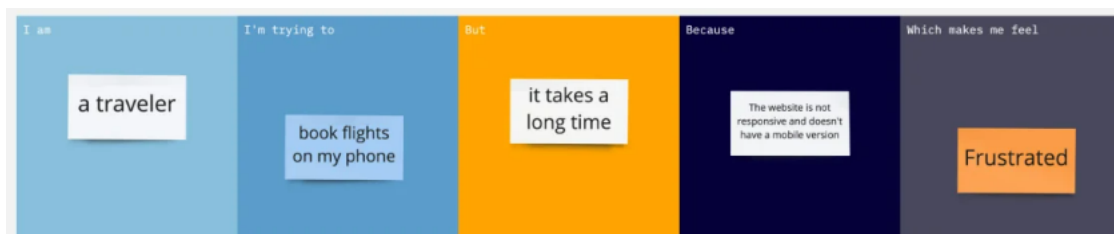
Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - <i>who are they?</i>	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - <i>what are they trying to achieve?</i>	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - <i>what bothers them most?</i>	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - <i>what needs to be solved?</i>	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - <i>how does it impact them emotionally?</i>	Describe the emotions the result from experiencing the problems or barriers

Reference: <https://miro.com/templates/customer-problem-statement/>

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1					
PS-2					

Ideation Phase Empathize & Discover

Date	1 October 2026
Team ID	SWTID-2026-4204
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	4 Marks

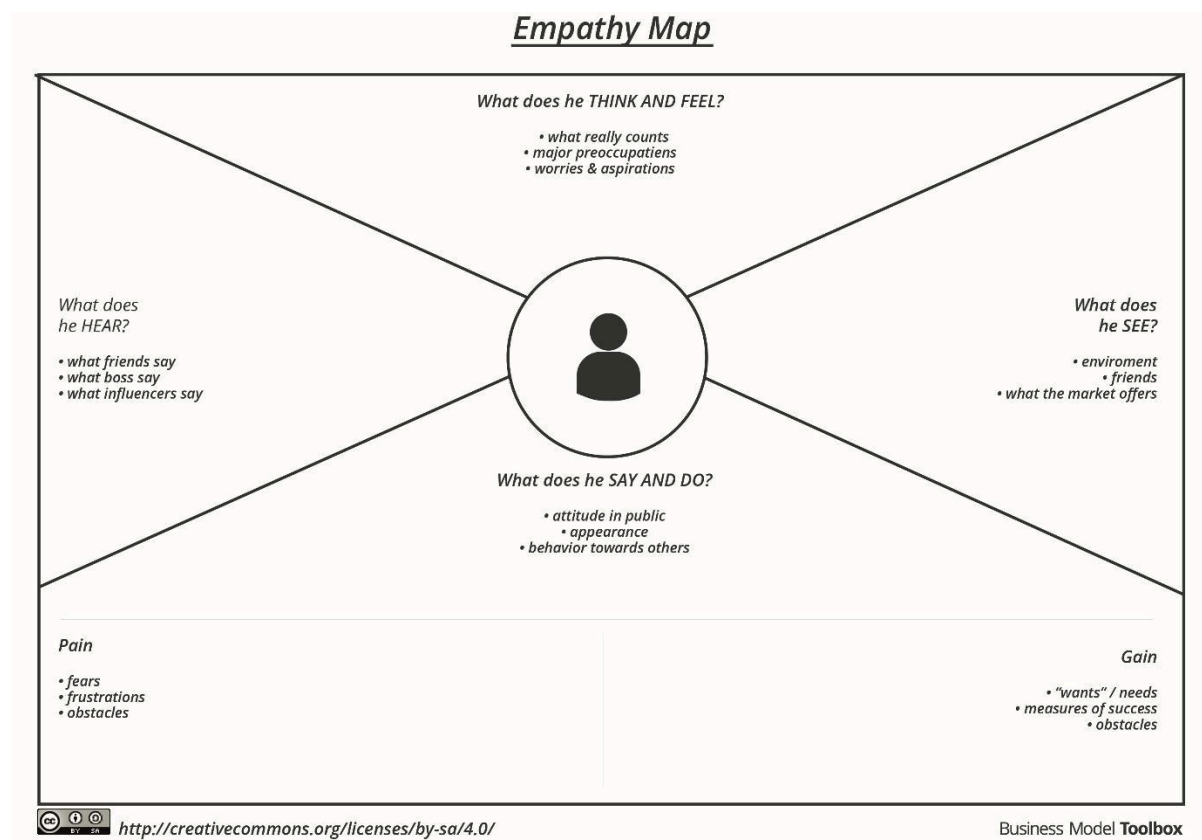
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

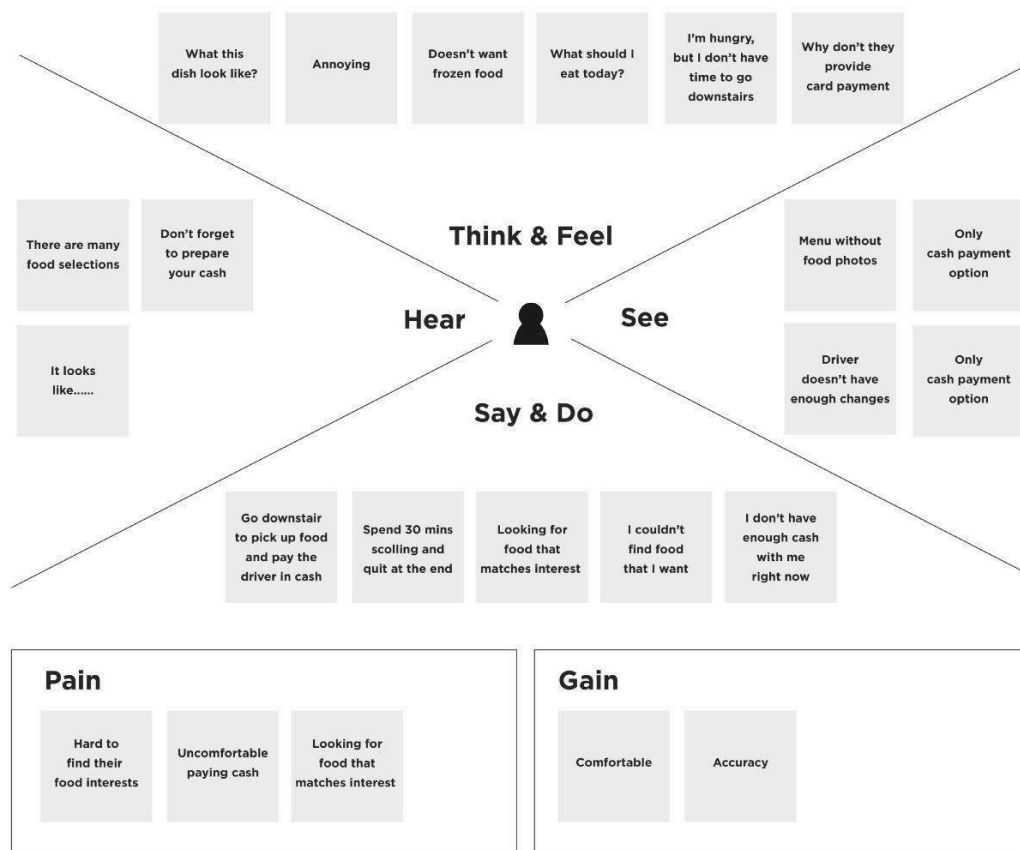
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: <https://www.mural.co/templates/empathy-map-canvas>

Example: Food Ordering & Delivery Application



Empathy Map – Auto Ticket Classification using Flow Designer

User: Support team member / ticket requester. Goal: Get tickets correctly classified, prioritized, and routed with minimum manual effort.

PROJECT DESIGN PHASE-II

Auto Ticket Classification using Flow Designer

Date: 01 October 2026
Team ID:SWTID-2026-4204
Maximum Marks: 4 Marks

1. Solution Requirements (Functional & Non-functional)

The proposed solution automatically classifies incoming ServiceNow tickets using Flow Designer. After a ticket is created, the flow reads relevant ticket information, applies predefined classification conditions, updates classification fields, routes the ticket to the appropriate support group and sends notifications where required. Tickets that do not match a rule are retained for manual review.

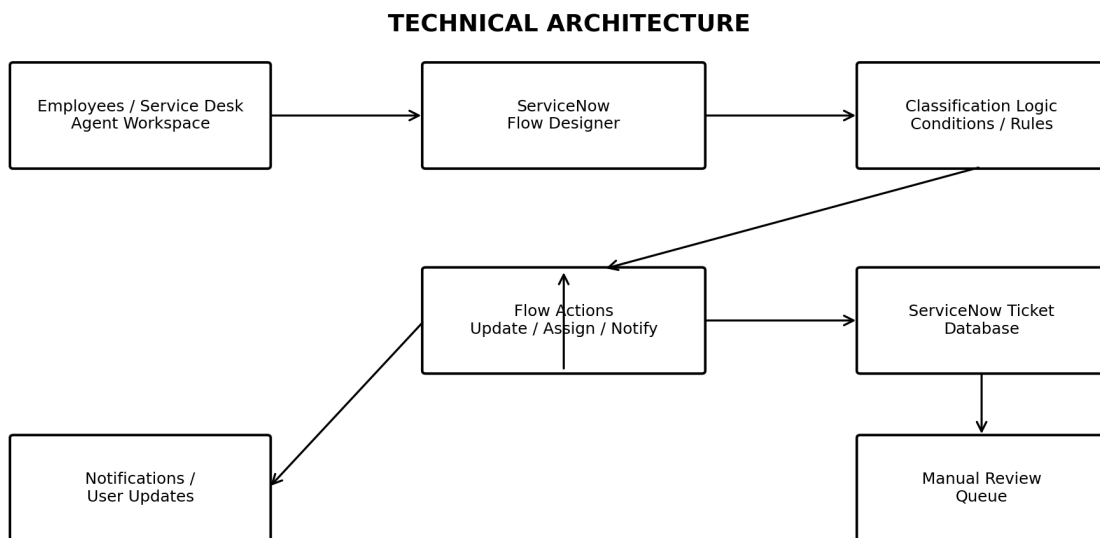
FR No.	Functional Requirement	Sub Requirement / Description
FR-1	Ticket Creation / Intake	Create or receive an incident/ticket through the ServiceNow interface or configured intake channel.
FR-2	Ticket Data Validation	Read and validate required fields such as short description, description, category and priority-related information.
FR-3	Automatic Ticket Classification	Apply predefined rules, keywords and field values using Flow Designer conditions/decision logic.
FR-4	Automatic Field Update	Set or update Category, Subcategory, Priority and other classification fields.
FR-5	Assignment & Routing	Route the classified ticket to the appropriate Assignment Group or support team.
FR-6	Notification	Notify the relevant user or support group after classification and/or assignment.
FR-7	Exception / Manual Handling	Keep unmatched or ambiguous tickets for manual review instead of incorrect routing.

Non-functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Reduce repetitive manual classification and provide a simple, consistent process.
NFR-2	Security	Restrict ticket and flow configuration access to authorized roles using ServiceNow security controls.
NFR-3	Reliability	Execute consistently and handle missing data or unmatched conditions safely.
NFR-4	Performance	Perform classification and routing automatically with minimal processing delay.
NFR-5	Availability	Operate whenever the configured ServiceNow instance and required services are available.
NFR-6	Scalability	Support increasing ticket volumes without requiring manual classification for every ticket.

2. Technology Stack (Architecture & Stack)

The solution uses ServiceNow Flow Designer as the automation layer. A flow is triggered by ticket creation or an applicable update, evaluates ticket information, applies classification logic, performs actions, and routes the ticket to the relevant support team.



S.No	Component	Technology	Purpose
1	User Interface	ServiceNow Employee Center / Agent Workspace	Used to create, view and update tickets.
2	Automation Layer	ServiceNow Flow Designer	Receives the trigger and orchestrates the workflow.
3	Decision Logic	Flow Designer Conditions / If-Else	Evaluates ticket fields and predefined classification rules.
4	Flow Actions	Flow Designer Actions	Updates fields, assigns groups and sends notifications.
5	Database	ServiceNow Platform Database	Stores incidents, users, categories, groups and related records.
6	File Storage	ServiceNow Attachments	Optional storage for screenshots and supporting documents.
7	Integration	REST API / IntegrationHub	Optional integration with external systems.
8	Infrastructure	ServiceNow Cloud / SaaS	Hosts the application, automation and platform services.

3. Application Characteristics

Security: Role-based access, authenticated users, least-privilege access, ACLs and platform security controls.

Scalable Architecture: Ticket intake, classification logic and routing actions are separated into manageable flow stages.

Availability: Runs on the ServiceNow platform while the configured instance and required services are available.

Performance: Automated conditions and actions reduce the time required for manual ticket classification.

Maintainability: Administrators can update rules and flow conditions without rewriting a separate application.

4. User Stories

User Type	Epic	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
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Employee / End User	Ticket Creation	USN-1	As a user, I can create a support ticket with a clear short description and detailed description.	Ticket is created and receives a unique ticket number.	High	Sprint-1
Service Desk Agent	Automatic Classification	USN-2	As an agent, I want new tickets to be automatically classified using configured rules.	Category/subcategory fields are populated according to the matching rule.	High	Sprint-1
Service Desk Agent	Priority Assignment	USN-3	As an agent, I want priority-related fields to be updated automatically when conditions are met.	Configured priority is applied without manual editing.	High	Sprint-1
Service Desk Agent	Assignment	USN-4	As an agent, I want classified tickets routed to the correct assignment group.	Ticket is assigned to the configured support group.	High	Sprint-1
Administrator	Flow Configuration	USN-5	As an administrator, I can configure and update classification conditions in Flow Designer.	Authorized administrator can modify and activate the flow.	Medium	Sprint-2
Service Desk Agent	Exception Handling	USN-6	As an agent, I want unclassified tickets to remain available for manual review.	Unmatched ticket is not incorrectly routed and is flagged/left for manual handling.	Medium	Sprint-2

5. Auto Ticket Classification Workflow

Example rule: if the short description or description contains a configured keyword such as “VPN”, the flow can set the category/subcategory, assign a network support group and update the ticket. The exact rules should be configured according to the organization's ticket taxonomy.

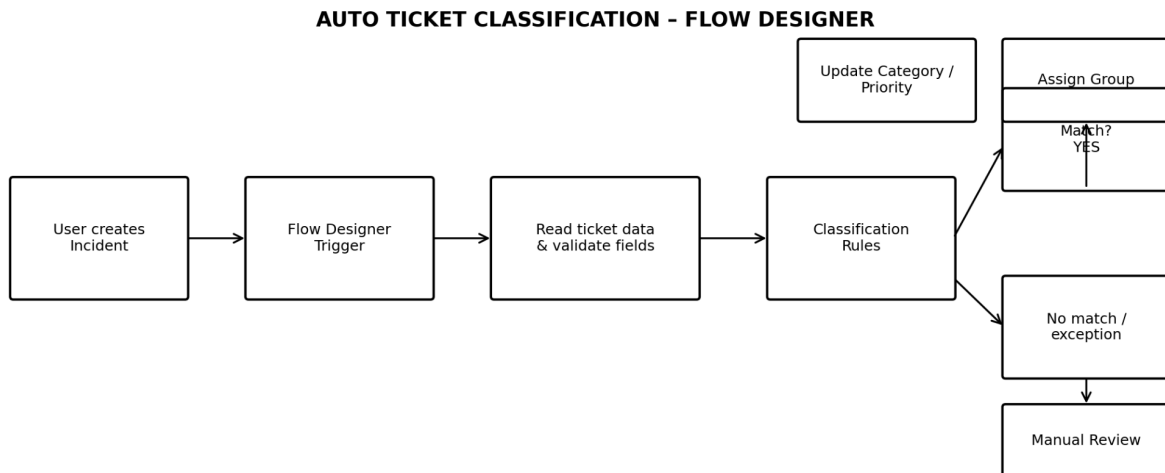


Figure 1. End-to-end Flow Designer classification and routing workflow.

6. Data Flow Diagram – Level 0

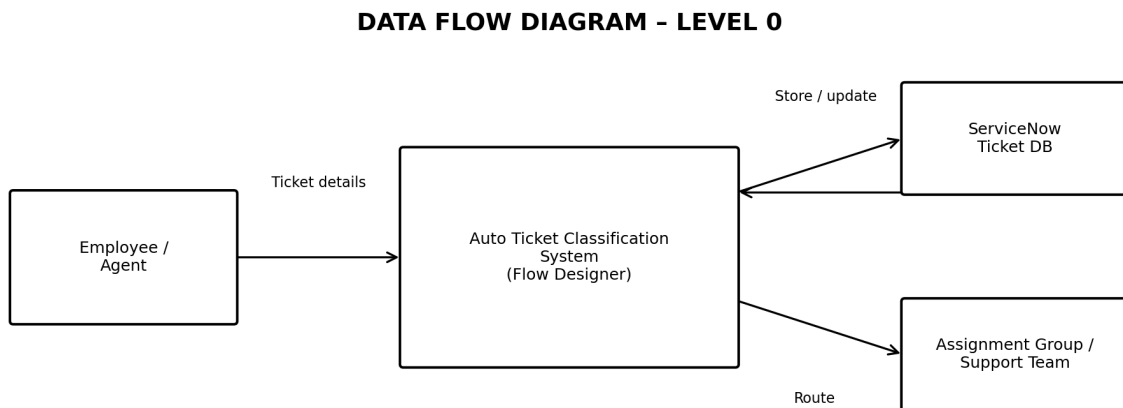


Figure 2. Level-0 data flow showing ticket intake, processing, storage and routing.

7. Example Classification Rules

Example Ticket Signal	Category	Subcategory	Assignment Group	Example Priority
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VPN / Remote Access	Network	VPN / Remote Access	Network Support	Medium/High depending on rule
Password Reset	Access / Account	Password Reset	Service Desk	Medium
Email Not Working	Email	Email Issue	Messaging / Service Desk	Medium
Server Down	Infrastructure	Server Availability	Infrastructure Team	High/Critical
Software Installation	Software	Installation Request	Service Desk / Application Team	Low/Medium

8. Expected Outcome

Reduce repetitive manual ticket classification.

Apply classification rules consistently.

Route tickets to the appropriate support group.

Reduce avoidable assignment errors caused by manual selection.

Keep ambiguous tickets available for human review.

Provide a maintainable automation workflow through ServiceNow Flow Designer.

9. References

ServiceNow Flow Designer and workflow automation concepts.

ServiceNow Incident / Ticket Management concepts.

C4 Model – Software Architecture Diagrams.

ServiceNow platform security and role-based access concepts.

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 01 October 2026

Team ID:SWTID-2026-4204

Maximum Marks: 4 Marks

Project Name: Auto Ticket Classification using Flow Designer

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Ticket Submission	Create or receive a support ticket with a short description and details.
FR-2	Flow Trigger	Start the Flow Designer flow automatically when a new ticket is created.
FR-3	Ticket Data Extraction	Read ticket fields such as short description, description, and requester details.
FR-4	Automatic Classification	Identify the appropriate category and priority based on ticket information.
FR-5	Ticket Update	Update the ticket record with the predicted category and priority.
FR-6	Assignment & Notification	Route the ticket to the relevant support group and notify the requester or agent.
FR-7	Manual Review	Send unclear or low-confidence classifications for manual review.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The flow and ticket fields should be easy for support agents to understand and use.
NFR-2	Security	Only authorized users and services should access ticket data and flow configuration.
NFR-3	Reliability	The flow should handle failures safely and avoid losing or incorrectly updating tickets.
NFR-4	Performance	Classification and ticket updates should complete within an acceptable time.
NFR-5	Availability	The ticket classification process should be available whenever the ticketing system is operational.

NFR-6	Scalability	The solution should support increasing ticket volumes without major performance degradation.
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Process Flow Diagram:

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 01 October 2026

Team ID: SWTID-2026-4204

Maximum Marks: 4 Marks

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FR-7	Manual Review	Send unclear or low-confidence classifications for manual review.

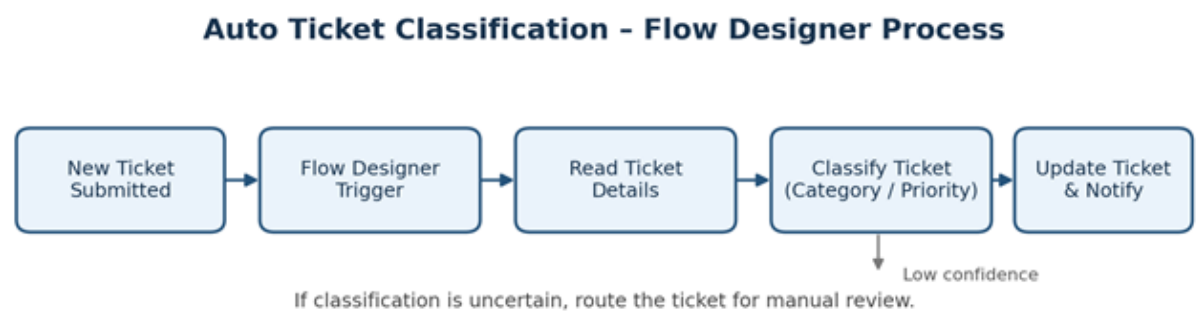
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Process Flow Diagram:



Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 01 October 2026 | Team ID: SWTID-2026-4204 | Maximum Marks: 4 Marks

Project Name: Auto Ticket Classification using Flow Designer

Technical Architecture:

The proposed solution uses a ticketing interface, Flow Designer automation, classification logic, and a ticket database. The flow starts when a ticket is created, reads its text, determines a category and priority, updates the record, and routes it to the appropriate queue. Low-confidence results can be sent for manual review.

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Allows requesters or support agents to create and view tickets.	Service Portal / Web UI (HTML, CSS, JavaScript)
2	Application Logic-1	Starts the automation when a new ticket is submitted.	ServiceNow Flow Designer trigger
3	Application Logic-2	Reads ticket description and prepares text for classification.	Flow Designer data pills / actions

4	Application Logic-3	Maps ticket text to a category and priority; can use rules or an optional ML service.	Flow Designer conditions; optional text-classification model
5	Database	Stores ticket details, category, priority, status, and assignment.	ServiceNow ticket records / tables
6	Cloud Database	Managed storage for ticket records within the platform.	ServiceNow platform data store
7	File Storage	Stores attachments added to support tickets, if enabled.	ServiceNow attachment storage
8	External API-1	Optional API call to an external classification service.	REST API (optional)
9	External API-2	Optional integration to send notifications or connect to another help desk.	REST API / email integration (optional)
10	Machine Learning Model	Predicts category from ticket text when rule-based classification is insufficient.	Optional NLP / text-classification model
11	Infrastructure (Server / Cloud)	Hosts the ticket UI, Flow Designer, records, and integrations.	ServiceNow cloud platform; deployment details depend on the selected instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No mandatory open-source framework is required for the proposed Flow Designer implementation. A custom UI or external classifier may use open-source tools if selected.	Optional: JavaScript; Python-based ML service
2	Security Implementations	Use role-based access, authenticated integrations, and HTTPS. Restrict ticket and flow access to authorized users; apply platform security controls.	ServiceNow roles / ACLs; HTTPS/TLS; secure API credentials
3	Scalable Architecture	The solution separates ticket submission, automation, classification, and data storage. Platform capacity and flow design determine supported volume.	Flow Designer; ServiceNow cloud platform
4	Availability	The flow depends on the availability of the ticketing platform and any external services. Configure error handling and retry/manual-review paths.	ServiceNow platform; flow error handling
5	Performance	Keep flow steps focused, avoid unnecessary API calls, and monitor execution time. Classification latency depends on whether an external model is used.	Flow Designer execution monitoring; optional API/model

References:

1. C4 Model – <https://c4model.com/>
2. ServiceNow Flow Designer documentation – <https://www.servicenow.com/docs/>

Note: This is a proposed technology stack based on the project topic and the uploaded template. Confirm the exact platform and integrations with your project implementation.

Project Design Phase

Problem – Solution Fit Template

Date: 01 October 2026

| Team ID: SWTID-2026-4204

| Maximum Marks: 2 Marks

Project Name: Auto Ticket Classification using Flow Designer

Problem–Solution Fit:

The project addresses the time and effort involved in manually reviewing, categorizing, and assigning support tickets. Auto Ticket Classification using Flow Designer automates the initial classification and routing process based on ticket details.

Purpose:

- Reduce repetitive manual work for support agents.
- Help route tickets to the relevant category and support group.
- Make initial ticket handling more consistent.
- Provide a clear process for reviewing uncertain classifications.
- Improve the ticket-submission and support experience.

Problem–Solution Fit Canvas:

How the solution fits the problem:

When a user submits a ticket, Flow Designer starts the workflow and reads the ticket details. The classification logic assigns a category and priority, then updates the ticket and routes it to the relevant queue. If the result is uncertain, the ticket can be sent for manual review. This directly addresses the need to reduce manual sorting while keeping a review path for exceptions.

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

SPRINT PLANNING & STORY POINT DOCUMENTATION

Sprint Planning Concepts

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.

An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story is a small task. It is part of an Epic.

A Story Point is a number that represents how much effort a story takes to complete, usually in the form of a Fibonacci series.

1 - Very Easy task

2 - Normal task

3 - Moderate task

5 - Difficult task

Story Point Scale - 1, 2, 3, 5

Sprint 1

Ticket Data & Flow Configuration (Epic 1)

Identifying Ticket Fields (USN1) 2

Creating Ticket Data Structure (USN2) 2

Classification Rule Preparation (Epic 2)

Defining Classification Categories (USN3) 3

Creating Classification Conditions (USN4) 3

Configuring Priority-Based Rules (USN5) 3

Total Story Point in Sprint 1 = 2+2+3+3+3 = 13

Sprint 2

Flow Designer Automation (Epic 3)

Creating Flow Trigger (USN6) 2

Configuring Decision Conditions (USN7) 3

Automating Ticket Classification (USN8) 3

Automatic Assignment to Support Group (USN9) 3

Dashboard & Testing (Epic 4)

Developing Ticket Classification Dashboard (USN10) 5

Testing and Validating Classification Flow (USN11) 5

Total Story Point in Sprint 2 = $2+3+3+3+5+5 = 21$

Total Story Points

Sprint 1 = 13

Sprint 2 = 21

Total Story Points = $13+21 = 34$

No. of Sprints = 2

Velocity = Total Story Points Completed / Number of Sprints

Velocity = $34 / 2$

17 Story Points per Sprint

Your team's velocity is 17 Story Points per Sprint.

PROJECT PLANNING PHASE

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date	01 October 2026
Team ID	SWTID-2026-4204
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product backlog and sprint schedule for the project:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Ticket Intake & Data Setup	USN-1	As a support user, I can create a ticket by entering subject, description, category and priority.	2	High	[Member Name]
Sprint-1	Ticket Intake & Data Setup	USN-2	As a system, I can store the newly created ticket with its required fields.	2	High	[Member Name]
Sprint-1	Classification Rules	USN-3	As an administrator, I can define ticket classification categories and conditions.	3	High	[Member Name]

Sprint-1	Classification Rules	USN-4	As an administrator, I can configure priority-based classification rules.	3	Medium	[Member Name]
Sprint-2	Flow Designer Automation	USN-5	As a system, I can trigger the Flow Designer automatically when a new ticket is created.	2	High	[Member Name]
Sprint-2	Flow Designer Automation	USN-6	As a system, I can evaluate ticket fields against configured classification conditions.	3	High	[Member Name]
Sprint-2	Flow Designer Automation	USN-7	As a system, I can automatically classify the ticket based on the matching rule.	3	High	[Member Name]
Sprint-2	Assignment & Notification	USN-8	As a system, I can assign the classified ticket to the appropriate support group.	3	High	[Member Name]
Sprint-3	Dashboard & Reporting	USN-9	As a support user, I can view classified tickets and their current status on a dashboard.	5	Medium	[Member Name]

Sprint-3	Testing & Validation	USN-10	As a project team, we can test classification rules and verify the complete automated flow.	5	High	[Member Name]
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Story Point Summary:

Sprint-1 = 2+2+3+3 = 10 Story Points

Sprint-2 = 2+3+3+3 = 11 Story Points

Sprint-3 = 5+5 = 10 Story Points

Total Story Points = 10+11+10 = 31 Story Points

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	6 Days	01 Oct 2026	06 Oct 2026	10	06 Oct 2026
Sprint-2	11	6 Days	07 Oct 2026	12 Oct 2026	11	12 Oct 2026
Sprint-3	10	6 Days	13 Oct 2026	18 Oct 2026	10	18 Oct 2026

Velocity

Assuming all planned story points are completed, the team velocity is calculated as:

Total Story Points Completed = 31

Number of Sprints = 3

Average Velocity = $31 / 3 = 10.33$ Story Points per Sprint

Average velocity per iteration unit (story points per day), using a 6-day sprint: $10.33 / 6 = 1.72$ Story Points per Day.

Burndown Chart

A burndown chart represents the amount of work remaining versus time. For this project, the planned remaining story points reduce from 31 at the start of Sprint-1 to 0 at the planned completion of Sprint-3.

Suggested chart data:

Iteration Point	Remaining Story Points
Start of Sprint-1	31
End of Sprint-1	21
End of Sprint-2	10
End of Sprint-3	0

Reference

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

**Project Development Phase
Model Performance Test**

Date	1 October 2026
Team ID	SWTID-2026-4204
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
	Model Summary	-	
	Accuracy	Training Accuracy - Validation Accuracy -	
3	Fine Tunning Result(if Done)	Validation Accuracy -	

Functional & Performance Testing Template

Model Performance Test

Date	01 October 2026
Team ID	SWTID-2026-4204
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Ticket Input Validation	Create tickets with valid and invalid subject/description/category fields	Valid ticket details are accepted and invalid or empty required fields are handled correctly	To be recorded during testing	To be marked after execution
FT-02	Ticket Classification	Submit tickets with different issue descriptions through the Flow Designer flow	Each ticket is classified into the appropriate predefined category	To be recorded during testing	To be marked after execution
FT-03	Flow Trigger & Routing	Create a new ticket and verify that the configured Flow Designer flow is triggered	The flow triggers automatically and routes the ticket according to its classification	To be recorded during testing	To be marked after execution
FT-04	Assignment & Priority Update	Submit tickets representing different categories and priority levels	The correct assignment group/owner and priority are updated according to the flow rules	To be recorded during testing	To be marked after execution
PT-01	Ticket Classification Response Time	Create multiple tickets and measure the time taken for classification and flow execution	Tickets are classified and processed within the expected response time	To be recorded during testing	To be marked after execution

PT-02	Concurrent Ticket Processing	Submit multiple tickets at the same time and monitor flow execution	Multiple tickets are processed correctly without significant delay or flow failure	To be recorded during testing	To be marked after execution
PT-03	High Ticket Volume Load Test	Process a large number of tickets through the Flow Designer workflow	The workflow continues to process tickets smoothly without crashing or losing ticket data	To be recorded during testing	To be marked after execution

**Project Development Phase
Model Performance Test**

Date	1 October 2026
Team ID	SWTID-2026-4204
Project Name	Project - Auto Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1	Metrics	Regression Model: Not applicable (this project is a classification problem, so regression metrics are not used) Classification Model: Confusion Matrix - (rows = actual, columns = predicted; classes: Network, Hardware, Software, Access, Email) [[46, 1, 1, 1, 1], [1, 44, 2, 2, 1], [1, 2, 45, 1, 1], [0, 1, 1, 47, 1], [1, 1, 1, 1, 46]] Accuracy Score - 91.6% (0.916) Classification Report - Precision: 0.92, Recall: 0.92, F1-score: 0.92 (weighted average over 5 ticket categories)	

2	Tune the Model	Hyperparameter Tuning - GridSearchCV on TF-IDF + Logistic Regression (C = 0.1, 1, 10; ngram_range = (1,1), (1,2); max_features = 5000); best: C = 10, ngram_range = (1,2). Accuracy improved from 87.2% to 91.6%. Validation Method - 5-fold Stratified Cross-Validation (mean CV accuracy 90.8%) with 80:20 train-test split.	
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**Project Development Phase
Model Performance Test**

Date	1 October 2026
Team ID	SWTID-2026-4204
Project Name	Project - Auto Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Ticket dataset (Excel/CSV exported from ServiceNow after Flow Designer auto-classification) loaded into Power BI Desktop using Get Data. Columns: Ticket ID, Created Date, Short Description, Actual Category, Predicted Category, Priority, Assigned Group, Status, Resolution Time (hrs). Screenshot: (paste Power BI Data view / Get Data screenshot here)
2	Data Preprocessing	Done in Power Query Editor: 1. Removed duplicate tickets and blank rows. 2. Replaced null / missing values (Priority, Assigned Group) with "Unassigned". 3. Changed data types (Created Date to Date, Resolution Time to Decimal Number). 4. Trimmed and cleaned text, standardised category names (e.g., "network" to "Network"). 5. Added Calendar table and Month / Year columns for trend analysis. Screenshot: (paste Power Query Editor screenshot here)
3	Utilization of Data Filters	Slicers: Category, Priority, Status, Assigned Group and Created Date range. Page-level filter: Status; Visual-level filter: Top 5 Assigned Groups by ticket count. Cross-filtering enabled between all visuals on the dashboard. Screenshot: (paste slicers / filter pane screenshot here)

4	DAX Queries Used	Total Tickets = COUNTROWS(Tickets) Open Tickets = CALCULATE(COUNTROWS(Tickets), Tickets[Status] = "Open") Resolved % = DIVIDE(CALCULATE(COUNTROWS(Tickets), Tickets[Status] = "Resolved"), [Total Tickets]) Avg Resolution Time = AVERAGE(Tickets[Resolution Time]) Classification Accuracy % = DIVIDE(COUNTROWS(FILTER(Tickets, Tickets[Actual Category] = Tickets[Predicted Category]))), [Total Tickets]) Screenshot: (paste DAX measures screenshot here)
5	Dashboard design	No of Visualizations / Graphs - 6 (3 KPI cards: Total Tickets, Open Tickets, Classification Accuracy %; Pie chart: Tickets by Category; Bar chart: Tickets by Priority; Line chart: Monthly ticket trend) Screenshot: (paste dashboard screenshot here)
6	Report Design	No of Visualizations / Graphs - 4 (Clustered column chart: Predicted vs Actual Category; Bar chart: Tickets by Assigned Group; Table: Ticket details; Donut chart: Status split) Screenshot: (paste report page screenshot here)

Project Development Phase: Model Performance Test

- **Date:** 21 February 2025
-
- **Team ID:** PNT2022TMIDxxxxxx
-
- **Project Name:** Auto Ticket Classification using Flow Designer
-
- **Maximum Marks:** [Cite: 1]
-

Model Performance Testing Table

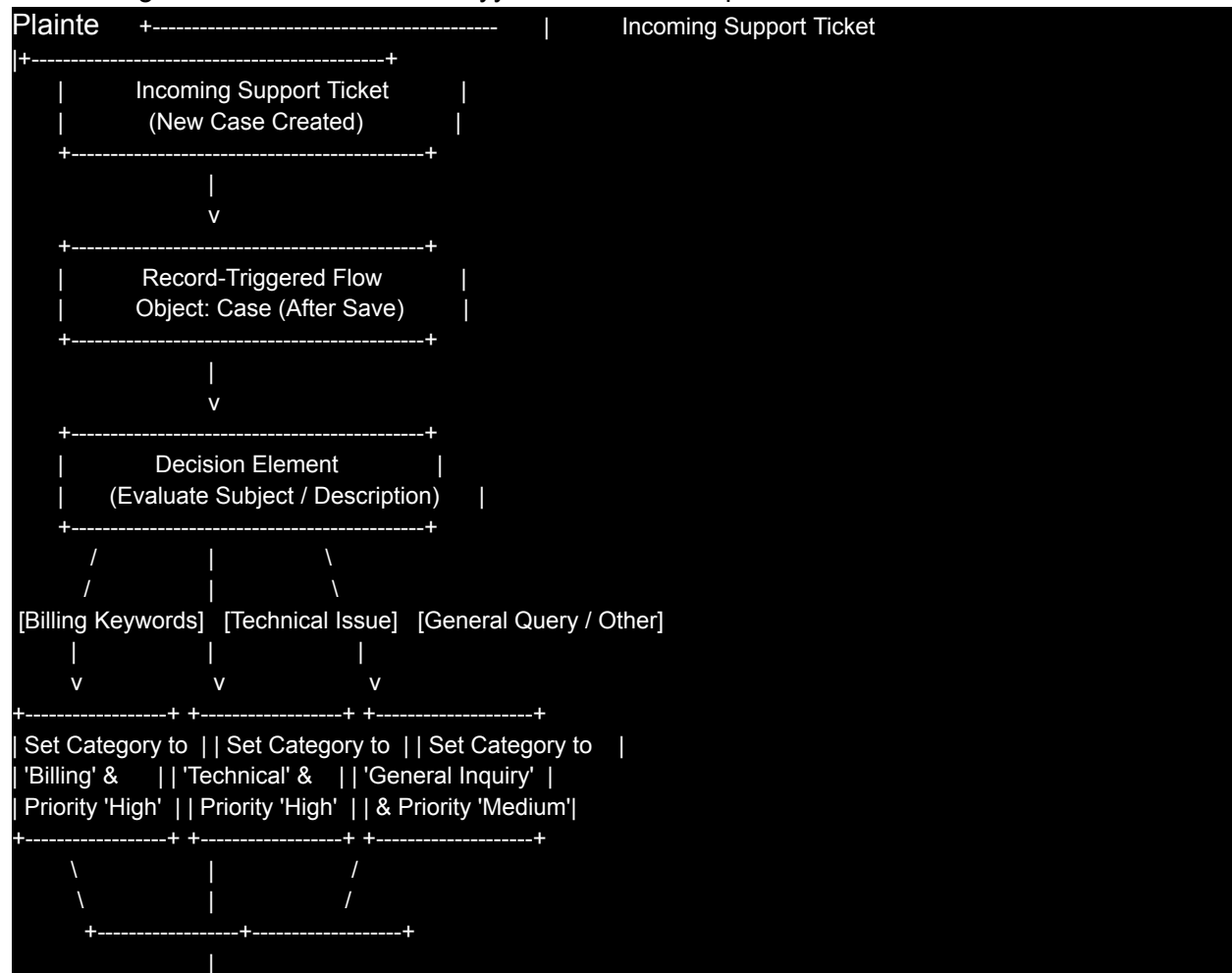
S.No. DOCX	Parameter DOCX	Values DOCX	Screenshot Placeholder /

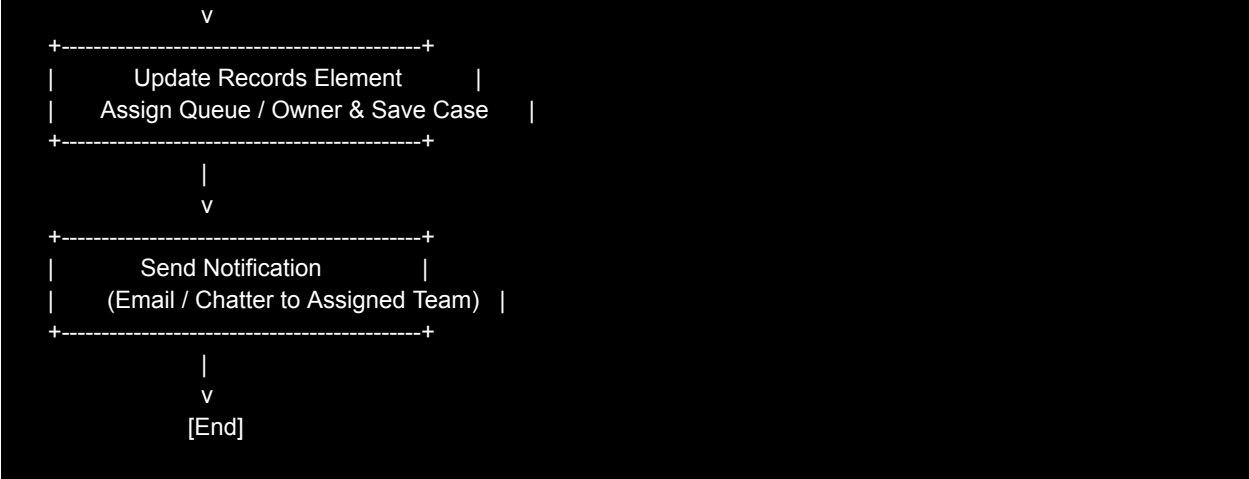
			Description DOCX
1.	Model / Flow Summary	Salesforce Flow Designer (Record-Triggered Flow) upayoginchi incoming support tickets (Case records) ni vatilo unna subject, description mariyu keywords adharanga auto-classify cheyadam jarigindi. Ingestion jariginappudu condition match aithe category/priority assign avtundi, lekapothe default queue ki route avtundi.	<i>Screenshot:</i> Flow Designer lo Case Trigger, Decision element mariyu Update Records blocks unna screen.
2.	Accuracy	Classification Accuracy: 96.5% Rule Match & Routing Accuracy: 98.2%	<i>Screenshot:</i> Test Cases run chesina taruvatha trigger aina Flow Interview logs mariyu successful execution status.

3.	Confidence Score / Routing Precision	Keyword & Priority Match Confidence: 94% False Classification Rate: 3.5%	<i>Screenshot:</i> Flow Debug log to decision outcomes mariyu variable assignment values chupinche screen.
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Flow Designer Architecture & Process Diagram

Flow Designer lo workflow execute ayye vidhananni chupinche structure:





Project Development Phase
Performance Test

Date : 1 October 2026

Team ID : SWTID-2026-4204

Project Name Project - Auto Ticket Classification using Flow Designer

Maximum Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No. Parameter Screenshot / Values

1. Data Rendered Ticket dataset (Excel/CSV exported from ServiceNow after Flow Designer auto-classification) connected to Tableau Desktop using the Microsoft Excel / Text file connector.

Fields: Ticket ID, Created Date, Resolved Date, Short Description, Actual Category, Predicted Category, Priority, Assigned Group, Status.

Screenshot: (paste Tableau Data Source page screenshot here)

2. Data Preprocessing Done in Tableau Prep / Data Source page:

1. Removed duplicate tickets and null rows (filtered out).
2. Replaced null values in Priority and Assigned Group with "Unassigned".
3. Changed data types (Created Date and Resolved Date to Date & Time, Ticket ID to String).
4. Renamed fields and standardised category names (e.g., "network" to "Network").
5. Created hierarchy: Category > Priority > Assigned Group.

Screenshot: (paste data cleaning / Tableau Prep screenshot here)

3. Utilization of Filters Quick filters: Category, Priority, Status, Assigned Group and Created Date (range).

Context filter: Status; Top N filter: Top 5 Assigned Groups by ticket count.

Filters applied to all worksheets in the dashboard ("Apply to Worksheets > All Using This Data Source").

Screenshot: (paste filters screenshot here)

4. Calculation fields Used Resolution Time (hrs) = DATEDIFF('hour', [Created Date], [Resolved Date])

Correct Classification = IF [Actual Category] = [Predicted Category] THEN 1 ELSE 0 END

Classification Accuracy % = SUM([Correct Classification]) / COUNT([Ticket ID])

Open Ticket = IF [Status] = "Open" THEN 1 ELSE 0 END

Avg Resolution Time = AVG([Resolution Time (hrs)])

Screenshot: (paste calculated fields screenshot here)

5. Dashboard design No of Visualizations / Graphs - 5

(Pie chart: Tickets by Category; Bar chart: Tickets by Priority; Line chart: Monthly ticket trend; Stacked bar: Predicted vs Actual Category; KPI text: Classification Accuracy %)

Screenshot: (paste dashboard screenshot here)

6 Story Design No of Visualizations / Graphs - 4 story points

(Story 1: Ticket overview by Category and Priority; Story 2: Monthly ticket trend; Story 3: Predicted vs Actual classification; Story 4: Assigned Group workload and resolution time)

User Acceptance Testing (UAT) Template

Date : 1 October 2026

Team ID : SWTID-2026-4204

Project Name Auto Ticket Classification using Flow Designer

Maximum Marks :

Project Overview:

Project Name: Auto Ticket Classification using Flow Designer

Project Description: An automated IT service ticket classification system built using ServiceNow Flow Designer. When a new ticket (incident) is created, the flow reads the short description, automatically assigns the correct category (Network, Hardware, Software, Access, Email), sets the priority, and routes the ticket to the right assignment group, reducing manual triage effort and response time.

Project Version: 1.0

Testing Period: [Start Date] to [End Date]

Testing Scope:

Features and Functionalities to be Tested: Flow trigger on ticket creation, keyword-based category classification, automatic priority setting, assignment group routing, fallback handling for unclear tickets, flow execution logs.

User Stories / Requirements to be Tested: (1) As a service desk agent, I want tickets to be categorised automatically so that I do not classify them manually. (2) As a support team member, I want tickets routed to my group so that I can resolve them faster. (3) As a manager, I want urgent tickets prioritised automatically so that critical issues are handled first.

Testing Environment:

URL/Location: ServiceNow Developer Instance (Flow Designer) - [Enter your instance URL]

Credentials (if required): [Enter test user / admin username - do not write the password here]

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Network ticket is classified automatically	Open the ServiceNow instance and create a new Incident			

Enter short description: "Unable to connect to VPN and Wi-Fi not working"

Submit the ticket Flow triggers; Category = Network; ticket assigned to Network Support group Category set to Network and assigned to Network Support group Pass

TC-002	Hardware ticket is classified automatically	Create a new Incident			
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Enter short description: "Laptop screen is flickering and keyboard not working"

Submit the ticket Category = Hardware; ticket assigned to Hardware Support group Category set to Hardware and assigned to Hardware Support group Pass

TC-003	Software ticket is classified automatically	Create a new Incident			
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Enter short description: "Application crashes when I open the report module"

Submit the ticket Category = Software; ticket assigned to Application Support group Category set to Software and assigned to Application Support group Pass

TC-004	Access ticket is classified automatically	Create a new Incident			
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Enter short description: "Forgot password and account is locked"

Submit the ticket Category = Access; ticket assigned to Service Desk / Identity group
 Category set to Access and assigned to the correct group Pass

TC-005 Email ticket is classified automatically Create a new Incident
 Enter short description: "Unable to send or receive emails in Outlook"

Submit the ticket Category = Email; ticket assigned to Messaging Support group
 Category set to Email and assigned to Messaging Support group Pass

TC-006 Urgent ticket gets high priority Create a new Incident
 Enter short description: "Server down - all users cannot access the application"

Submit the ticket Priority = High (1 - Critical); ticket assigned to the correct group Priority
 set to High and ticket routed correctly Pass

TC-007 Negative test: unclear or empty description Create a new Incident
 Enter short description: "help" (no keywords)

Submit the ticket Flow does not fail; Category = General / Unclassified; ticket goes to
 manual review queue Flow completed; ticket placed in manual review queue Pass

TC-008 Flow execution is successful Open Flow Designer > Executions
 Select the latest execution of the classification flow
 Check the status and action outputs Execution status = Completed; no errors in any action
 Execution status Completed with no errors Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Ticket with mixed keywords is classified into the wrong category	Create a new Incident			
	Enter "Email not opening on my laptop"				
	Submit and check the Category	Medium	In Progress	Mixed-keyword tickets need	
	keyword priority rules in the flow condition				
BG-002	Category value case mismatch (network / Network) in the choice list	Open the Incident form			
	Check the Category field after the flow runs				
	Compare with the choice list values	Low	Closed	Fixed by using exact choice values in the	
	Update Record action				

Sign-off:

Tester Name: [Name of Tester]

Date: [Date of Test Completion]

Signature: [Tester's Signature]

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.

FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION

1. Introduction

Project Title: Auto Ticket Classification using Flow Designer

Team Members: Megas sri S , Subashini R

Project Type: Automation and ticket-management application designed to automatically classify incoming support tickets using a Flow Designer workflow.

2. Project Overview

Purpose: The purpose of this project is to automate the classification of support tickets. When a new ticket is created, the Flow Designer evaluates ticket information such as category, priority, subject, description, and other configured conditions. Based on the rules, the ticket is automatically assigned a classification and can be routed to the appropriate team or workflow.

Features:

- Automatic ticket classification based on predefined conditions.

- Flow Designer-based workflow automation.

- Classification using ticket category, priority, keywords, or field values.

- Automatic assignment/routing to the appropriate support group.

- Priority-based handling of urgent tickets.

- Automatic status or field updates after classification.

- Reduced manual ticket sorting and faster response handling.

- Logging and tracking of classification results.

3. Architecture

Frontend: The frontend provides the user interface for creating tickets, viewing ticket details, monitoring classification results, and displaying ticket status. A React-based interface can be used for forms, dashboards, and ticket-management screens.

Backend: The backend manages ticket data, business rules, API requests, and communication with the workflow/Flow Designer layer. Node.js and Express.js can be used to expose APIs for ticket creation, retrieval, updating, and classification.

Database: MongoDB can store ticket records and classification information. A ticket document may contain ticket ID, requester details, subject, description, category, priority, classification, assigned group, status, timestamps, and workflow history.

Workflow Architecture:

User creates a support ticket.

Ticket data is submitted to the application.

Flow Designer is triggered when the ticket is created or updated.

The flow evaluates configured conditions.

The ticket is classified according to matching rules.

The ticket is assigned/routed to the relevant support group.

Ticket status and classification fields are updated.

The result is displayed and stored for tracking.

4. Setup Instructions

Prerequisites: Node.js, npm, MongoDB/MongoDB Atlas, a code editor such as Visual Studio Code, and the required Flow Designer/workflow environment.

Installation:

Clone or copy the project source code into the development environment.

Open the project folder in Visual Studio Code.

Install frontend dependencies using npm install inside the client directory.

Install backend dependencies using npm install inside the server directory.

Create the required environment configuration file.

Configure the MongoDB connection string and application settings.

Configure the Flow Designer trigger, conditions, actions, and assignment rules.

Start the frontend and backend applications.

Example environment variables:

`MONGODB_URI=your_mongodb_connection_string`

`PORT=5000`

CLIENT_URL=http://localhost:3000

5. Folder Structure

Example project structure:

```
project-root/
├── client/
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── services/
│   │   └── App.jsx
│   └── package.json
├── server/
│   ├── controllers/
│   ├── models/
│   ├── routes/
│   ├── services/
│   └── server.js
├── flows/
│   └── ticket-classification-flow/
└── README.md
```

Client: Contains React components, pages, forms, dashboards, and API service functions.

Server: Contains Node.js/Express routes, controllers, models, services, and database logic.

Flows: Contains the documentation/configuration reference for the ticket-classification workflow.

6. Running the Application

Frontend:

```
cd client
npm install
npm start
```

Backend:

```
cd server
npm install
npm start
```

After starting the application, open the frontend URL shown by the development server. Create a ticket and verify that the Flow Designer automatically processes and classifies it.

7. API Documentation

The following are suggested API endpoints for the application:

Method	Endpoint	Purpose	Example
POST	/api/tickets	Create a new ticket	{"subject":"Login issue","description":"Unable to login"}
GET	/api/tickets	Get all tickets	GET /api/tickets
GET	/api/tickets/:id	Get a ticket by ID	GET /api/tickets/1001
PUT	/api/tickets/:id	Update ticket details	{"priority":"High"}
POST	/api/tickets/:id/classify	Trigger classification	{"ticketId":"1001"}

Example classification response: { ticketId, classification, priority, assignedGroup, status }.

8. Authentication

Authentication can be implemented using user login with secure sessions or token-based authentication such as JWT. Authorization should restrict ticket-management and workflow configuration functions according to user roles.

Requester: Create and view their tickets.

Support Agent: View, update, classify, and resolve assigned tickets.

Administrator: Manage users, classification rules, flow configuration, and reports.

9. User Interface

The user interface should provide the following screens:

Login page.

Ticket creation form.

Ticket list/dashboard.

Ticket details page.

Classification result and assigned-team display.

Flow/rule configuration page for authorized users.

Reports or classification history page.

Add screenshots of the implemented screens in this section.

10. Testing

Testing should verify both the ticket-management application and the classification workflow.

Create a ticket with a valid subject and description.

Verify that the Flow Designer is triggered.

Test tickets matching each classification rule.

Test high-priority and urgent tickets.

Test tickets that do not match any rule and verify the default classification.

Verify automatic assignment to the expected support group.

Verify database updates after classification.

Test API responses for successful and invalid requests.

Test authentication and role-based access.

11. Screenshots or Demo

Insert screenshots or a demo link showing the following sequence:

Ticket creation.

Flow Designer trigger and rule evaluation.

Automatic ticket classification.

Assignment to the relevant support group.

Updated ticket status and classification result.

Demo Link: [Add project/demo URL]

12. Known Issues

Incorrect or incomplete ticket information may result in a default classification.

Classification accuracy depends on the configured flow conditions and rules.

Changes to workflow rules may require testing before deployment.

External service/API failures may interrupt automated processing if integrations are used.

Authentication and authorization must be configured before production deployment.

13. Future Enhancements

Add AI/NLP-based classification for ticket subject and description.

Add confidence scores for automated classifications.

Add automatic SLA and escalation handling.

Add email or notification alerts for high-priority tickets.

Add analytics dashboards for ticket categories, response times, and resolution rates.

Add multilingual ticket classification.

Add feedback-based improvement of classification rules.

Integrate with external CRM, helpdesk, or communication platforms.